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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. Each part is sorted independently
c. The parts are ignored
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Quick Sort
c. Bubble Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **4**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It sorts data using a single pass
c. It is less efficient than bubble sort
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It makes the analysis slower
c. It has no effect on the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It divides the list into two halves, sorts each half, and then merges them
c. It compares adjacent elements and swaps them if necessary
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **7**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Decreases the efficiency of algorithms
- c. Makes code execution slower
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Finding a specific element in a list
- c. Removing duplicates from a list
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting helps in finding duplicates quickly
c. Sorting makes it harder to search for items
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Bubble Sort
b. Binary Search
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sorted()
- c. sort()
- d. arrange()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Merge Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Merge Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Bubble Sort